



UNIVERSITI PUTRA MALAYSIA

“BERILMU BERBAKTI”

CCS3700 – EMBEDDED SYSTEMS PROGRAMMING

LAB 4

LED CHASE EFFECT

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Part 1 LED Chase Effect

In this lab exercise, students will attempt to build a circuit (refer Figure 1) that enables the LED chase effect. We are now going to use a string of LED's (10 in total) to make an LED chase effect, like that used on the car KITT in the Knight rider TV Series and on the way introduce the concept of arrays. **An array is a collection of variables that are accessed using an index number.** In this lab exercise, we shall explore the approach of using arrays in Arduino programming.

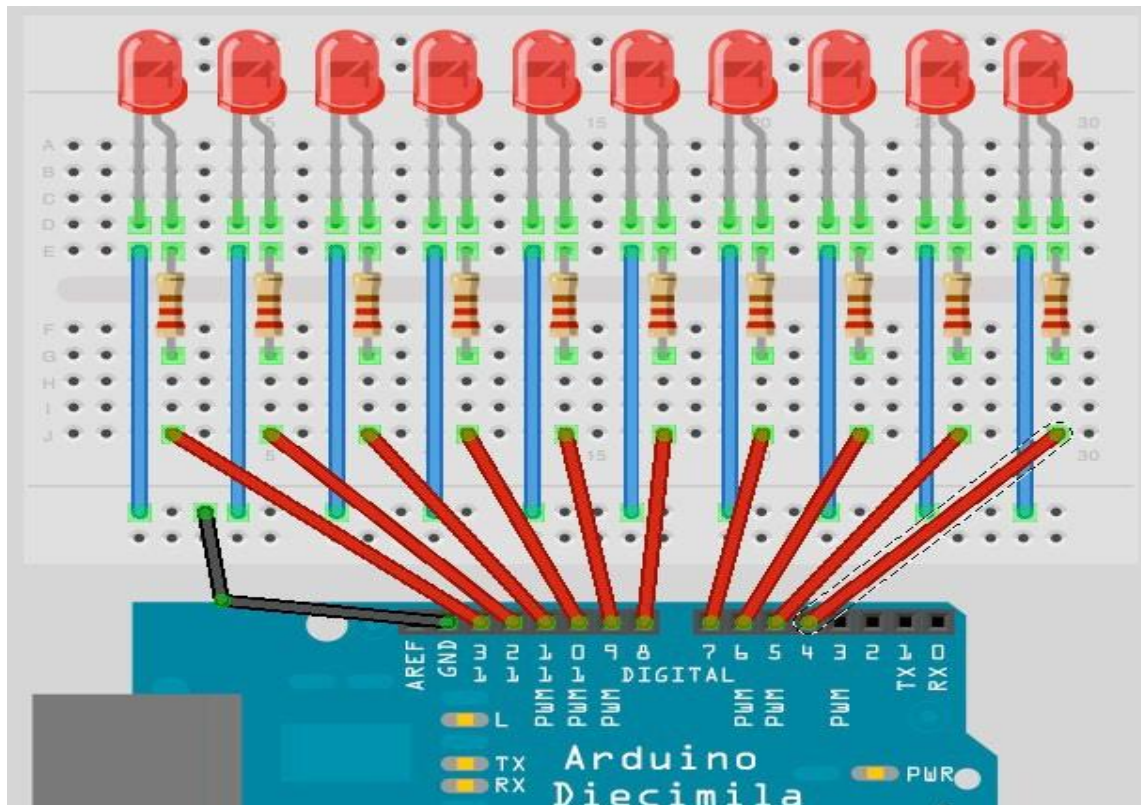


Figure 1: LED Chase Effect

Hardware

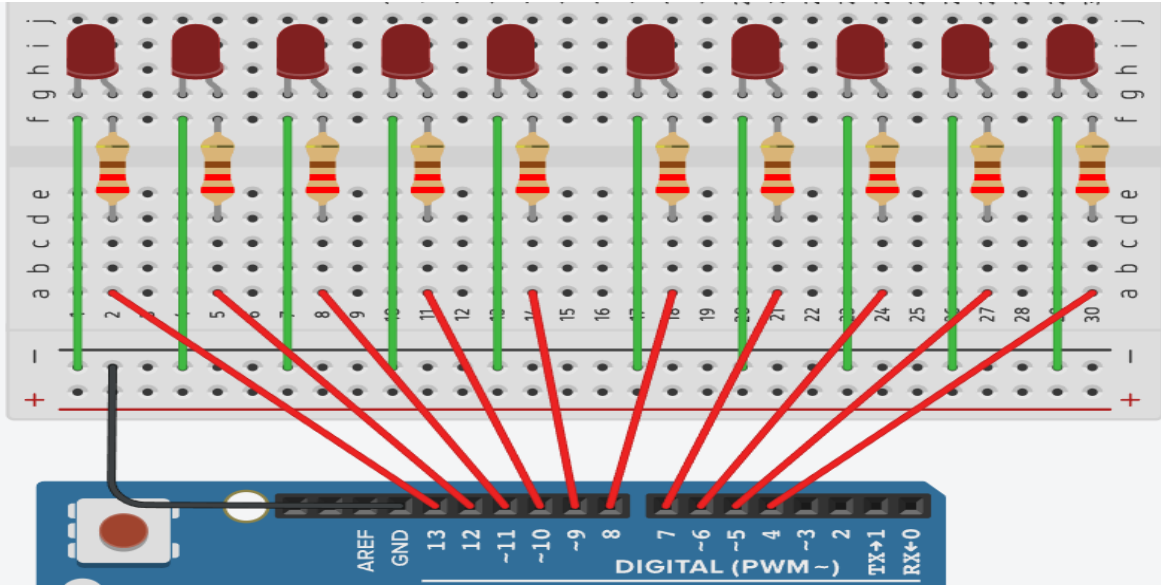
10 x red LEDs
10 x 220Ω resistors
Breadboard
Jumper wires

Instruction

Write the required program and compile it into your Arduino board. Your first line of your sketch may begin as follows:

```
byte ledPin[] = {4, 5, 6, 7, 8, 9, 10, 11, 12, 13};
```

THE LAYOUT



ENTIRE CODE

```
// Define the array of LED pins
byte ledPin[] = {4, 5, 6, 7, 8, 9, 10, 11, 12, 13};

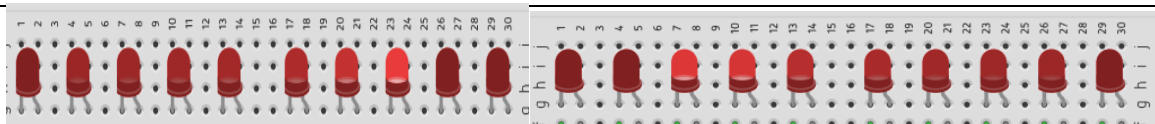
// Set the delay time (in milliseconds) to control the speed of the chase
int ledDelay = 65;

void setup() {
  // Use a 'for' loop to iterate through the array and set each pin as an OUTPUT
  for (int i = 0; i < 10; i++) {
    pinMode(ledPin[i], OUTPUT);
  }
}

void loop() {
  // Forward sweep: From the first LED (index 0) to the last (index 9)
  for (int i = 0; i < 10; i++) {
    digitalWrite(ledPin[i], HIGH); // Turn the current LED on
    delay(ledDelay);               // Wait for the specified delay time
    digitalWrite(ledPin[i], LOW);  // Turn the current LED off
  }

  // Backward sweep: From the second-to-last LED (index 8) to the second (index 1)
  // We start at 8 and end at 1 so the LEDs at the far ends don't flash twice in a row
  for (int i = 8; i > 0; i--) {
    digitalWrite(ledPin[i], HIGH); // Turn the current LED on
    delay(ledDelay);               // Wait for the specified delay time
    digitalWrite(ledPin[i], LOW);  // Turn the current LED off
  }
}
```

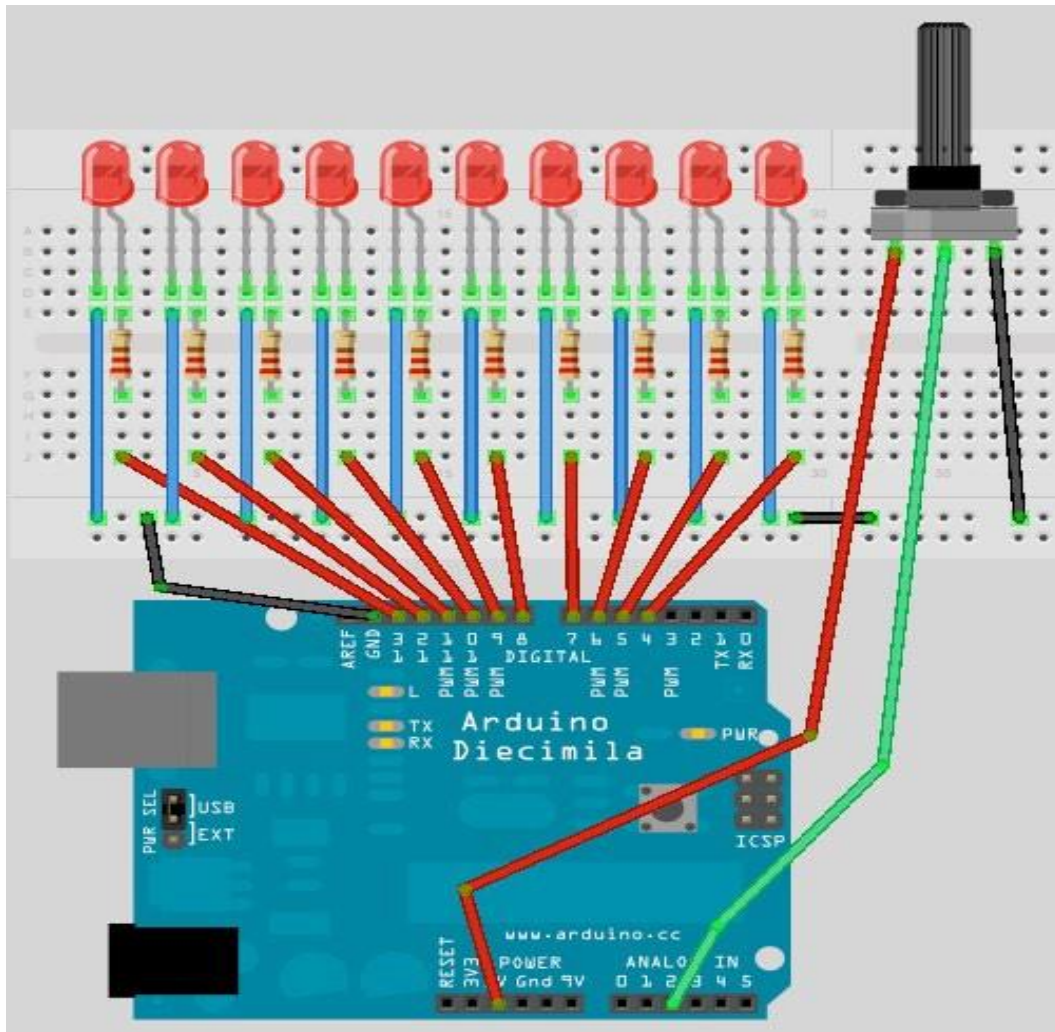
OUTPUT



IT ROTATES TO BOTH SIDES ENDLESSLY

Part 2 Interactive LED Chase Effect

In this lab exercise, students would need to use the same circuit with the addition of a potentiometer. Refer to Figure 2 below for the configuration.

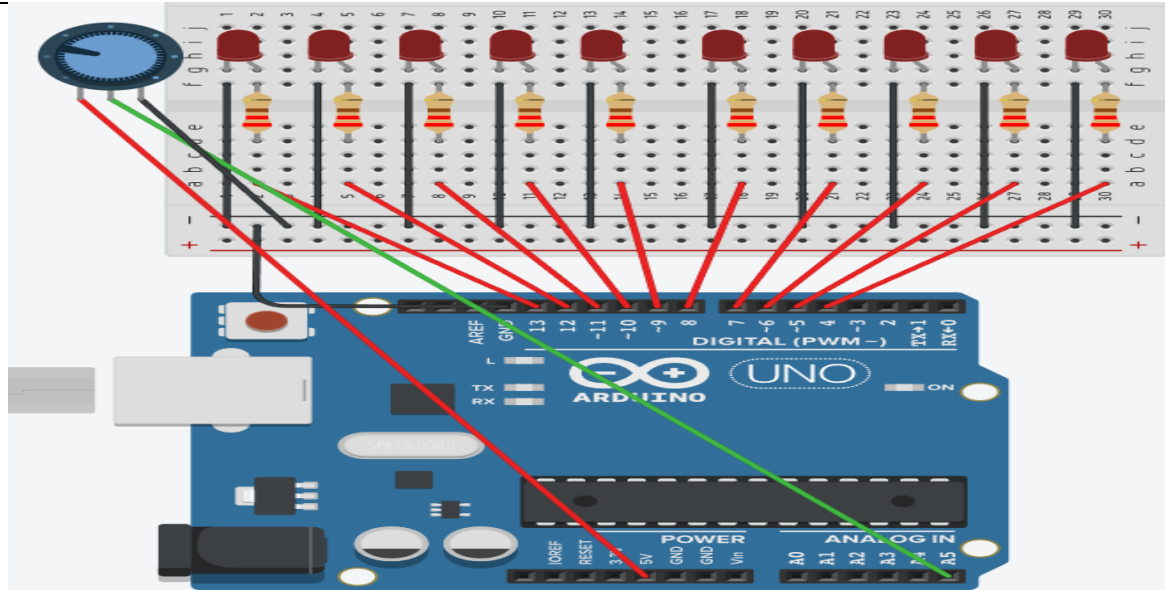


Instruction

Write the required program and compile to Arduino for the results. This is the same circuit used in Part 1, but with added potentiometer that is connected to 5V, Ground and analog Pin 5.

After verifying and uploading the code, you should see the lit LED appear to bounce back and forth between each end of the string of the lights as before. But, by turning the knob of the potentiometer, you will change the value of delay and speed up or slow down the effect.

THE LAYOUT



THE CODE

```
// Define the array of LED pins
byte ledPin[] = {4, 5, 6, 7, 8, 9, 10, 11, 12, 13};

// Define the analog pin connected to the potentiometer
int potPin = 5; // Refers to Analog In 5 (A5)

// Variable to store the delay time, which will be read from the potentiometer
int ledDelay;

void setup() {
  // Set each pin in the array as an OUTPUT
  for (int i = 0; i < 10; i++) {
    pinMode(ledPin[i], OUTPUT);
  }
}

void loop() {
  // Read the potentiometer value (0 to 1023) and set it as the delay time
  ledDelay = analogRead(potPin);

  // Forward sweep: From index 0 to 9
  for (int i = 0; i < 10; i++) {
    // We read the potentiometer inside the loops as well so it responds instantly
    ledDelay = analogRead(potPin);

    digitalWrite(ledPin[i], HIGH);
    delay(ledDelay);
    digitalWrite(ledPin[i], LOW);
  }

  // Backward sweep: From index 8 down to 1
  for (int i = 8; i > 0; i--) {
    // Read the potentiometer value again for instant response
    ledDelay = analogRead(potPin);

    digitalWrite(ledPin[i], HIGH);
    delay(ledDelay);
    digitalWrite(ledPin[i], LOW);
  }
}
```

THE OUTPUT



THE SPEED FOLLOWS THE POTENTIOMETER, THE HIGHER IT IS, THE FASTER IT IS